

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 02:17 UTC

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- Cycles recorded: **215**
- Time span: 124.6h (~1.73 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	192
FALSIFIED	14
BARRIER_HIT	5
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Random Matrix Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1

Field	Cycles
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-28 17:25 UTC	INCONCLUSIVE	Hereditary Discrepancy of Clause-Variable Incidence Lower-Bounds
2026-04-28 17:55 UTC	INCONCLUSIVE	Schur-Polynomial Positivity Defect Lower-Bounds Tree-Frege Lines
2026-04-28 18:31 UTC	INCONCLUSIVE	Erdős-Rado Sunflower Petals Lower-Bound Lifted Log-Rank for IND_2
2026-04-28 19:01 UTC	INCONCLUSIVE	Subword Complexity of Lifted Rows Lower-Bounds Real Rank for IND_2
2026-04-28 19:42 UTC	INCONCLUSIVE	Bakry-Émery Curvature Floor Lower-Bounds Tseitin Resolution Width
2026-04-28 20:08 UTC	INCONCLUSIVE	Dyck Zero-Crossing Count of Row-Walks Capped by Log-Rank
2026-04-28 20:35 UTC	INCONCLUSIVE	Cyclotomic Norm Floor of Truth-Table Character Sum Caps $\text{ACC}^0[m]$
2026-04-28 23:06 UTC	INCONCLUSIVE	Hilbert-Compression Floor on Property-A Gadgets Forces $\alpha \cdot Q(f)$ Mul
2026-04-28 23:09 UTC	INCONCLUSIVE	Effective-Resistance Diameter Lower-Bounds Tseitin Resolution Width
2026-04-29 00:11 UTC	INCONCLUSIVE	$\text{GF}(2)$ -Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone

Time	Verdict	Title
2026-04-29 00:24 UTC	INCONCLUSIVE	Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity A
2026-04-29 00:40 UTC	INCONCLUSIVE	Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real
2026-04-29 01:14 UTC	INCONCLUSIVE	Secant Rank Lower Bound for Disjointness Communication Matrices
2026-04-29 01:46 UTC	INCONCLUSIVE	Sign-Rank Distinguishing Tensor for Read-Twice Branching Programs
2026-04-29 02:17 UTC	INCONCLUSIVE	SOS Hierarchy Integrality Gap for Random CSPs

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```